

TABLE 1 Summary of features and experimental observations for selected multicargo T3SC

Multicargo chaperone	pI/10 ³ M _w	Known binding interaction(s) with secreted substrates ^a	Mutant virulence defect	Interaction with conserved T3SS component	Relevant references
InvB	4.5/14.9	SspA (SipA), SopE/E2, SopA	Yes		6, 19, 20, 43
Spa15	4.3/15.2	IpaA, IpgB1, IpgB2, OspB, OspC1, OspC2, OspC3, OspD1, OspD2	ND ^b		60, 61, 67
CesT	4.4/17.7	Tir, Map, EspF, EspG, EspH, EspZ, NleA, NleG, NleH, NleH2	Yes	EscN	1, 16, 23, 73
HpaB (162 aa)	4.3/18.5	AvrBs1, AvrBs3, XopF1	Yes	HrcN, HrcU, HrcV	3, 7, 8, 46
Mcsc	4.6/18.8	Ct618, Cap1, Ct225	ND	CdsQ	68
SrcA	4.6/16.1	SseL	Yes	SsaN	11

^a Listed by direct interactions in experimental binding assays. Additional roles in effector secretion without demonstrated binding have been observed in some cases.

^b ND, not determined.

regardless of gene context. Within the last 10 years, a number of multicargo chaperones have been implicated in binding, recruitment, regulatory, and docking roles. The importance of these proteins is validated by findings where multicargo chaperone mutants exhibit significant virulence defects in animal and plant models of infection. The features of some of these chaperones are presented below and summarized in Table 1.

InvB. The *invABC* genetic locus located within the *Salmonella* pathogenicity island 1 (SPI-1) was initially implicated in eukaryotic cell invasion, although *invB* mutants were still able to invade cells *in vitro*, making it difficult to assess the contribution of InvB to pathogenesis (22). *invB* mutants were shown to have a virulence defect in an avian model of *Salmonella* infection, yet the exact contribution or functional role of InvB in virulence remained unknown (63). At the sequence level, InvB shares 31% sequence similarity with Spa15 (SpaK), a protein encoded in the *mxl-spa* genetic locus associated with *Shigella* host cell invasion, and therefore a link to *Salmonella* invasion was suggested. The first demonstration for InvB being a chaperone was a study that used a yeast two-hybrid approach and *in vitro* infection experiments. Bronstein et al. (6) discovered that InvB interacts with SspA (SipA) and contributes to efficient translocation of SipA into target host cells. Two groups then answered a key question relating to *Salmonella* effectors encoded outside SPI-1 by demonstrating that InvB was required for SopE and SopE2 secretion and translocation into host cells (19, 43). Finally, using a polymutant effector *Salmonella* strain, InvB was shown to interact with SopA (20). Aside from SopE and SopE2, the InvB interacting effectors do not share extensive sequence similarity, which raised the important question of how InvB mediates binding specificity to effector partner proteins. As it turns out, intrinsic structural features within effectors are critical for InvB binding (45) (presented below).

Spa15. The invasive enteric pathogen *Shigella flexneri* (and close relatives) contains a large 213-kb virulence plasmid that encodes components of a T3SS, effectors, chaperones, and transcriptional regulators. A screen for interacting proteins encoded by the virulence plasmid identified a number of potential chaperone-effector associations (58). Based on homology to InvB, which was previously known as a *Salmonella* chaperone for the effector SipA/SspA, a chaperone role for Spa15 was postulated based on an interaction with the effectors IpaA, IpgB1, and OspC3 (58, 60). This was the first evidence that T3SC could interact with multiple effectors and hence introduced the concept of “promiscuous” chaperones, i.e., chaperones that interact with multiple binding partners (35). In fact, a protein interaction platform assay has since

revealed that Spa15 interacts with additional *Shigella* effectors, including IpgB2, OspB, OspC1, OspC2, OspD1, and OspD2 (for a total of nine) (67). Spa15 is involved in the efficient secretion of all nine of these effectors. Interestingly, the OspD1-Spa15 interaction has been implicated in an anti-activator role that negatively influences virulence gene transcription (61). Another observation is that to date Spa15 is the only example of a T3SC that is secreted (26). The relevance of Spa15 secretion remains to be determined.

CesT. CesT of enteropathogenic *E. coli* (EPEC) is perhaps one of the most unusual multicargo chaperones, as it was first described as a class IA chaperone for the effector Tir (1, 23). CesT (chaperone *E. coli* secreted protein Tir) fits many of the criteria for assignment as a class IA chaperone; the *cesT* gene is immediately adjacent to *tir* within an operon in the locus of enterocyte effacement (LEE) pathogenicity island of attaching and effacing *E. coli* strains. This is similar to the cognate chaperone effector pairings for class IA chaperones seen in many pathogens (e.g., SycE/YopE, SigE/SigD, SicP/SptP). Additional interaction studies revealed that CesT binds to Map, another LEE-encoded effector (16). The discovery of many non-LEE-encoded effectors, accompanied by the paucity of chaperone candidate genes, raised the possibility that CesT could be involved in non-LEE-encoded effector secretion via the T3SS. Indeed this was the case, as CesT was shown to interact with all the effectors encoded within the LEE and many non-LEE-encoded effectors (73). CesT is known to interact with at least 10 effectors (Table 1) and possibly more given that some effectors are encoded by redundant genes with almost identical sequences (e.g., NleG and EspF protein families). Expectedly, *cesT* null mutants do not cause overt disease in a mouse model of infection (18).

HpaB. Plant pathogens within the genus *Xanthomonas* use a T3SS to inject multiple effectors into plant cells during infection. The model organism for this group of bacteria is *Xanthomonas campestris* pv. *vesicatoria*, which causes bacterial spot disease in pepper and tomato. Studies have identified a 23-kb hypersensitive response and pathogenicity (*hrp*) gene cluster that contains six operons (*hrpA-F*) encoding components of a T3SS, effectors, regulators and chaperones (27). Genes without known functions or limited homology to known T3SS proteins in databases were initially termed *hrp*-associated genes (*hpa*). In the case of HpaB, it was shown to bind three effector proteins, AvrBs1, AvrBs3, and XopF1 using *in vitro* binding assays (7, 9). *In vitro* secretion assays demonstrated that five *Xanthomonas* effectors require HpaB for efficient secretion into culture supernatants. HpaB is required for *Xanthomonas*-mediated T3SS injection of two effectors into plant